Applying Slot Attention to Past Tense

Alan Xie, Henry Zhang, Jacob Liao

March 2026

1 Introduction

The acquisition of the English past tense is a central test case for theories of linguistic generalization. This phenomenon is interesting because it combines a highly productive regular pattern with a persistent set of irregular exceptions, and because children's past-tense acquisition follows a characteristic U-shaped developmental trajectory: early correct production of a small set of high-frequency forms, followed by overregularization, and finally mastery of a large vocabulary of both regular and irregular verbs. As such, the past tense occupies privileged place in debates over whether language learning is best understood in terms of symbolic rules or distributed statistical learning. Our project proposes to use slot attention, a method that maps input features to latent vectors or "slots" that compete to explain the input, to learn present-tense to past-tense transduction.

2 Lit Review

The English past tense has often been used as a test case for whether neural networks can learn something like an abstract grammatical rule. Rumelhart and McClelland's classic model showed that it was possible to "teach" a neural network the past tense under a specific training regimen, in which irregular verbs were introduced early and regular patterns emerged later; under those conditions, the model reproduced the famous U-shaped learning curve associated with child language acquisition. However, that result has long been criticized as something of a "nice trick," since the behavior depends strongly on the order and structure of training and does not clearly match how humans actually acquire language. More recent work revisited the question with modern architectures. Kirov and Cotterell trained encoder-decoder RNNs on stem/past-tense pairs and evaluated them on both seen and nonce verbs, arguing that these models extend past-tense behavior in a relatively human-like way. But Corkery, Matusevych, and Goldwater re-evaluated that claim and found that the model's nonce-verb behavior did not match human judgments nearly as well as suggested, with weak and unstable human-model correspondence across simulations. Transformer-based follow-ups reached a similar conclusion. Ma and Gao showed that transformer performance depends heavily on the training setup, especially the type and token distributions of regular and irregular verbs, and although the models learn some aspects of regularity, they still fit human nonce-verb data poorly. Taken together, these results suggest a recurring pattern: training conditions may matter more than architecture, which makes it difficult to argue that these models are truly learning an abstract past-tense rule in the way humans do.

More recent language-modeling work has tried to move closer to the child-learning setting. Haga et al. trained small language models on small corpora and evaluated them not by direct present-to-past transduction, but by asking whether the model assigned a higher probability to a sentence with the correct inflection. They found some human-like behavior, including partial U-shaped patterns for some verbs, but not uniformly across the system; performance continued to oscillate in ways that suggest the models were not consistently learning a stable past-tense rule. This motivates our proposed use of Slot Attention. Rather than forcing one large vector to represent the entire input at once, Slot Attention breaks the input into a small number of reusable latent features. Although originally developed for images, it has been adapted to language by giving each slot its own learnable mean, fixing the initialization variance, and combining the slot bottleneck with a sparsification layer and a small Transformer decoder. Applied to past-tense learning, this architecture may allow the model to discover latent components of a word, such as the verb stem, a suffix pattern, or a phonological context, which could support cleaner generalization and more closely approximate what learning a "rule" would look like. This defines the goal of our project: to determine how well Slot Attention can learn past-tense rules.

3 Planned Methods

We propose to model past-tense learning as a supervised character-level transduction task from a present-tense input form to its corresponding past-tense output form. Given an input verb $x = (x_1, \ldots, x_n)$, a Transformer encoder will first map the character sequence to contextualized character representations $H \in R^{n \times d}$. These representations will then be passed to a Slot Attention module, which will iteratively compress the sequence into a fixed upper bound of K latent slot vectors $S = (s_1, \ldots, s_K)$. Because the slots attend competitively over the encoded character features, this stage will provide a structured bottleneck over the input representation. Following the architecture in the original slot-based sequence model, we will then apply an L_0-based sparsification layer (L0Drop) to the slot set, yielding a pruned memory $M' = \{m_1, \ldots, m_{K'}\}$ with $K' \leq K$. This will allow the model to use a variable number of active slots depending on the input while discouraging degenerate solutions in which each character is simply passed through independently.

The pruned slot memory M' will serve as the conditioning representation for an autoregressive character-level decoder that generates the target past-tense form $y = (y_1, \ldots, y_m)$. At each decoding step t, the decoder will predict the next output character conditioned on the previously generated target prefix and the slot memory:

$$p(y \mid x) = \prod_{t=1}^{m} p(y_t \mid y_{<t}, M').$$

In contrast to the original autoencoding setup, which reconstructs the input sequence itself, our proposed model will instead decode a distinct target sequence corresponding to the past-tense form. For example, the model will be trained to map *play* to *played*, *try* to *tried*, and *go* to *went*.

Training will be performed on present–past verb pairs using teacher forcing. The primary objective will be the negative log-likelihood of the target output sequence:

$$\mathcal{L}_{\text{transduce}} = -\sum_{t=1}^{m} \log p(y_t \mid y_{<t}, M').$$

To preserve sparsity in the induced slot representation, this loss will be combined with the L_0 regularization term associated with the L0Drop layer:

$$\mathcal{L} = \mathcal{L}_{\text{transduce}} + \lambda \mathcal{L}_{L_0},$$

where λ controls the strength of the sparsity penalty. We will also evaluate a multi-task variant in which the same pruned slot memory is shared by two decoders: a main decoder for present-to-past transduction and an auxiliary decoder for present-to-present reconstruction. Under this formulation, the full objective will be

$$\mathcal{L} = \mathcal{L}_{\text{transduce}} + \alpha \mathcal{L}_{\text{recon}} + \lambda \mathcal{L}_{L_0},$$

where $\mathcal{L}_{\text{recon}}$ denotes the autoregressive reconstruction loss on the original input sequence and α determines its weight relative to the main transduction objective.

4 Proposed Timeline

Week 1: Data Collection/Setup

- Collect and preprocess a dataset of English verb stem/past-tense pairs (regular and irregular verbs).
- Generate or compile a set of nonce verbs for evaluation.
- Setup environment if necessary.

Responsibility: all

Week 2: Model Implementation

- Implement Transformer encoder + Slot Attention model.
- Train initial model on dataset and verify correctness.

Responsibility: all

Week 3: Experiments and Evaluation

- Train model under different training regimes.
- Evaluate performance on seen verbs, unseen verbs, and nonce verbs.
- Conduct error analysis.

Responsibility: all

Week 4: Analysis and Write-Up

- Analyze results.
- Create tables and visualizations.

Responsibility: all

Week 5: Prep Presentation

- Write final report and prepare presentation.

Responsibility: all

5 Bibliography

Behjati, M., Henderson, J. (n.d.). Inducing Meaningful Units from Character Sequences with Dynamic Capacity Slot Attention. Beser, D. (2021). Falling Through the Gaps: Neural Architectures as Models of Morphological Rule Learning (arXiv:2105.03710). arXiv. https://doi.org/10.48550/arXiv.2105.03710

Corkery, M., Matusevych, Y., Goldwater, S. (2019). Are we there yet? Encoder-decoder neural networks as cognitive models of English past tense inflection. In A. Korhonen, D. Traum, L. Màrquez (Eds.), Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (pp. 3868–3877). Association for Computational Linguistics. https://doi.org/10.18653/v1/P19-1376

Haga, A., Sugawara, S., Fukatsu, A., Oba, M., Ouchi, H., Watanabe, T., Oseki, Y. (2024). Modeling Overregularization in Children with Small Language Models. In L.-W. Ku, A. Martins, V. Srikumar (Eds.), Findings of the Association for Computational Linguistics: ACL 2024 (pp. 14532–14550). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.findings-acl.865

Kirov, C., Cotterell, R. (2018). Recurrent Neural Networks in Linguistic Theory: Revisiting Pinker and Prince (1988) and the Past Tense Debate. Transactions of the Association for Computational Linguistics, 6, 651–665. https://doi.org/10.1162/tacl$_{a0}$0247

Locatello, F., Weissenborn, D., Unterthiner, T., Mahendran, A., Heigold, G., Uszkoreit, J., Dosovitskiy, A., Kipf, T. (2020, June 26). Object-Centric Learning with Slot Attention. arXiv.Org. https://arxiv.org/abs/2006.15055v2

Ma, X., Gao, L. (2022). How do we get there? Evaluating transformer neural networks as cognitive models for English past tense inflection. In Y. He,

H. Ji, S. Li, Y. Liu, C.-H. Chang (Eds.), Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 1: Long Papers) (pp. 1101–1114). Association for Computational Linguistics. https://doi.org/10.18653/v1/2022.aacl-main.81